

Figure 8: 7 models consistently exhibit more sycophancy in First-pov than in Third-pov , with an average increase of 13.6%, on all expertise levels (full result at Figure 14 in the Appendix).

appears to reduce the pressure to conform, allowing the model’s internal knowledge to influence its final output more freely.

Where Does the Model Encode the Pronoun Effect?

To find mechanistic evidence for the behavioral differences observed, we investigated where the model’s representations diverge under these different person framings. Specifically, using layer-wise KL divergence, we measured the difference between the hidden state distributions of First-pov and Third-pov prompts against a Plain baseline.

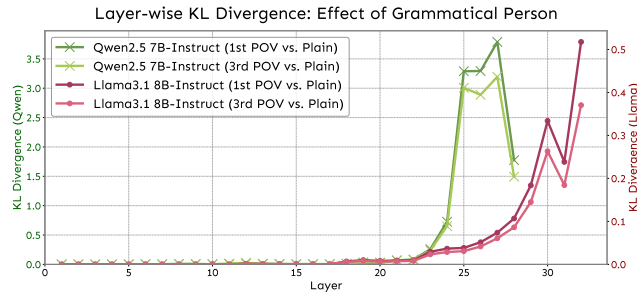


Figure 9: First-pov opinions induce an earlier and more significant representational shift in the model’s final layers compared to Third-pov opinions.

Figure 9 presents the divergence curves for both 1st- and 3rd-person conditions, using *Llama3.1 8B-Instruct* as a representative model. We observe that both conditions are processed similarly in the lower and middle layers, with KL divergence remaining negligible until approximately Layer 24. However, in the deeper layers, a sharp distinction emerges. While both framings cause the model’s representations to diverge from Plain , the First-pov forces a more dramatic shift, increasing more rapidly and reaching a substantially higher peak in the final layer.

To further clarify the nature of these representational differences, we analyzed the cosine similarity between category centroids at the critical layers identified by peak KL divergence (layer 32 for *Llama*, layer 27 for *Qwen*). Figure 10 visualizes PCA-projected hidden states from layer 32 of

Cosine Similarity between Category Centroids across 1st- and 3rd-person Prompts

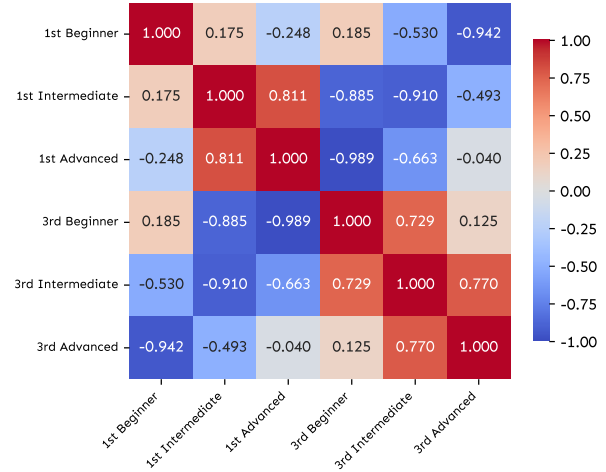


Figure 10: Cosine similarity heatmap between category centroids obtained from PCA-projected hidden states at layer 32 of *Llama*. Each centroid represents a specific combination of pronoun perspective (1st vs. 3rd person) and expertise level (Beginner, Intermediate, Advanced). *Qwen* result is in Figure 15, Appendix.

Llama, showing that pronoun framing leads to substantial angular separation in the model’s internal representations.

Notably, Figure 10 reveals that within-pronoun comparisons (upper-left and lower-right blocks) show high similarity values, with representations clustered by pronoun frame regardless of expertise level. Conversely, cross-pronoun comparisons (lower-left and upper-right blocks) display much lower or even negative cosine similarities (e.g., -0.04 between *1st Advanced* and *3rd Advanced*), indicating the model encodes first- and third-person prompts as nearly orthogonal directions in its representational space. These findings demonstrate that grammatical person creates distinct representational clusters in the model’s latent space, suggesting pronoun framing fundamentally alters how the model processes user opinions.

Takeaway 4

Grammatical person is a key driver of sycophancy in LLMs. Changing prompts from first- to third-person framing substantially reduces sycophantic behavior, with this effect encoded deep within the model’s representations. Our analysis confirms that grammatical person constitutes a more salient processing axis than expertise-level for how opinions are processed.

Conclusion

This study offers a mechanistic explanation for sycophancy in LLMs, showing it is opinion- rather than authority-driven, as they fail to represent authority internally. User opinions suppress learned knowledge in later layers, validated by causal activation patching. We also identify a strong perspective-driven effect: first-person prompts elicit more sycophancy than third-person ones by creating a stronger override of the model’s internal knowledge.

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